# GILLAM, MB, Canada

WMO: **719120**

Lat: **56.360N** Lon: **94.710W** Elev: **145** StdP: **99.59** Time Zone: **-6.00 (NAC)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-36.9** | **-34.9** | **-41.2** | **0.1** | **-36.7** | **-39.4** | **0.1** | **-34.7** | **10.9** | **-19.8** | **9.5** | **-20.2** | **3.5** | **270** | **0.790** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **11.2** | **27.3** | **17.8** | **25.4** | **16.9** | **23.6** | **16.2** | **19.5** | **24.7** | **18.2** | **23.1** | **17.2** | **22.0** | **4.5** | **220** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **17.7** | **12.9** | **21.7** | **16.3** | **11.8** | **20.4** | **15.1** | **10.9** | **19.5** | **56.5** | **24.7** | **52.3** | **23.1** | **48.9** | **22.2** | **28.4** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **10.7** | **9.4** | **8.4** | DB | **-39.9** | **31.7** | **2.2** | **2.0** | **-41.6** | **33.2** | **-42.9** | **34.3** | **-44.1** | **35.5** | **-45.7** | **37.0** | (4) |
| (5) | | | | WB | **-40.0** | **22.3** | **2.2** | **1.9** | **-41.6** | **23.7** | **-42.9** | **24.8** | **-44.1** | **25.9** | **-45.7** | **27.3** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **-3.1** | **-23.8** | **-21.4** | **-14.3** | **-4.4** | **3.9** | **12.2** | **16.2** | **14.6** | **8.8** | **0.7** | **-11.0** | **-20.0** | (6) |
| (7) | | DBStd | | **15.32** | **7.27** | **6.91** | **7.92** | **7.41** | **5.76** | **4.82** | **3.62** | **3.64** | **4.41** | **4.45** | **7.02** | **7.94** | (7) |
| (8) | | HDD10.0 | | **5279** | **1048** | **879** | **753** | **434** | **204** | **31** | **2** | **4** | **74** | **289** | **629** | **931** | (8) |
| (9) | | HDD18.3 | | **7863** | **1306** | **1113** | **1012** | **683** | **448** | **191** | **84** | **126** | **286** | **546** | **879** | **1189** | (9) |
| (10) | | CDD10.0 | | **496** | **0** | **0** | **0** | **1** | **16** | **97** | **195** | **147** | **39** | **1** | **0** | **0** | (10) |
| (11) | | CDD18.3 | | **39** | **0** | **0** | **0** | **0** | **1** | **7** | **19** | **11** | **1** | **0** | **0** | **0** | (11) |
| (12) | | CDH23.3 | | **455** | **0** | **0** | **0** | **0** | **12** | **110** | **210** | **109** | **13** | **1** | **0** | **0** | (12) |
| (13) | | CDH26.7 | | **88** | **0** | **0** | **0** | **0** | **1** | **22** | **42** | **22** | **2** | **0** | **0** | **0** | (13) |
| (14) | Wind | WSAvg | | **4.1** | **3.8** | **3.9** | **3.9** | **4.0** | **4.2** | **4.0** | **3.9** | **4.2** | **4.5** | **4.7** | **4.2** | **3.8** | (14) |
| (15) | Precipitation | PrecAvg | | **518** | **21** | **17** | **23** | **31** | **48** | **56** | **86** | **77** | **57** | **41** | **35** | **25** | (15) |
| (16) | | PrecMax | | **719** | **39** | **36** | **41** | **70** | **136** | **128** | **163** | **127** | **94** | **105** | **81** | **46** | (16) |
| (17) | | PrecMin | | **352** | **6** | **6** | **4** | **5** | **6** | **14** | **31** | **19** | **32** | **8** | **12** | **8** | (17) |
| (18) | | PrecStd | | **108** | **10** | **7** | **10** | **16** | **30** | **28** | **39** | **34** | **20** | **24** | **20** | **12** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **-2.0** | **0.7** | **7.1** | **18.0** | **25.3** | **29.3** | **30.6** | **29.4** | **25.1** | **15.4** | **4.5** | **-0.8** | (19) |
| (20) | | | MCWB | | **-2.5** | **-0.3** | **3.2** | **9.8** | **14.7** | **18.5** | **20.6** | **19.6** | **17.2** | **9.6** | **2.6** | **-1.4** | (20) |
| (21) | | 2% | DB | | **-6.0** | **-3.5** | **4.0** | **12.4** | **20.8** | **26.4** | **27.5** | **26.1** | **21.6** | **11.8** | **1.8** | **-3.7** | (21) |
| (22) | | | MCWB | | **-6.5** | **-4.6** | **1.1** | **6.1** | **12.2** | **16.2** | **18.0** | **18.0** | **15.4** | **8.5** | **0.7** | **-4.2** | (22) |
| (23) | | 5% | DB | | **-9.0** | **-7.2** | **1.1** | **9.1** | **17.1** | **24.0** | **25.6** | **24.0** | **18.5** | **9.0** | **0.2** | **-6.0** | (23) |
| (24) | | | MCWB | | **-9.5** | **-8.0** | **-0.6** | **4.3** | **10.3** | **15.2** | **17.3** | **17.1** | **13.8** | **6.8** | **-0.4** | **-6.4** | (24) |
| (25) | | 10% | DB | | **-12.4** | **-10.5** | **-1.8** | **6.1** | **13.9** | **21.6** | **23.8** | **21.6** | **15.8** | **6.8** | **-1.3** | **-8.0** | (25) |
| (26) | | | MCWB | | **-12.8** | **-11.2** | **-3.3** | **2.4** | **8.4** | **14.3** | **16.7** | **16.0** | **12.3** | **5.1** | **-1.9** | **-8.4** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **-2.5** | **-0.1** | **3.4** | **10.2** | **15.2** | **21.3** | **21.8** | **20.8** | **18.4** | **11.1** | **2.9** | **-1.4** | (27) |
| (28) | | | MCDB | | **-2.0** | **0.8** | **6.5** | **17.7** | **22.4** | **26.5** | **28.3** | **26.5** | **22.9** | **14.1** | **4.1** | **-1.0** | (28) |
| (29) | | 2% | WB | | **-6.5** | **-4.6** | **1.3** | **6.7** | **12.9** | **18.0** | **19.6** | **19.2** | **16.1** | **8.8** | **0.9** | **-4.1** | (29) |
| (30) | | | MCDB | | **-6.0** | **-3.5** | **3.6** | **11.6** | **19.7** | **23.5** | **24.8** | **24.2** | **20.7** | **11.4** | **1.7** | **-3.7** | (30) |
| (31) | | 5% | WB | | **-9.4** | **-8.1** | **-0.6** | **4.5** | **10.8** | **16.3** | **18.4** | **18.0** | **14.2** | **6.9** | **-0.4** | **-6.4** | (31) |
| (32) | | | MCDB | | **-9.0** | **-7.2** | **1.1** | **8.7** | **16.2** | **22.2** | **23.4** | **22.5** | **17.9** | **8.9** | **0.2** | **-6.0** | (32) |
| (33) | | 10% | WB | | **-12.8** | **-11.2** | **-3.4** | **2.5** | **8.8** | **14.8** | **17.4** | **16.7** | **12.7** | **5.2** | **-1.8** | **-8.3** | (33) |
| (34) | | | MCDB | | **-12.4** | **-10.4** | **-1.9** | **5.8** | **13.6** | **20.5** | **22.4** | **20.7** | **15.6** | **6.6** | **-1.3** | **-7.9** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **8.1** | **10.4** | **12.2** | **12.1** | **11.2** | **12.2** | **11.2** | **9.9** | **8.6** | **6.0** | **7.0** | **7.6** | (35) |
| (36) | | 5% DB | MCDBR | | **9.1** | **12.1** | **12.7** | **14.5** | **17.0** | **15.7** | **14.3** | **13.8** | **13.4** | **10.5** | **6.4** | **8.0** | (36) |
| (37) | | | MCWBR | | **9.0** | **11.3** | **10.3** | **9.4** | **9.8** | **7.7** | **7.0** | **7.4** | **7.9** | **7.4** | **5.8** | **7.9** | (37) |
| (38) | | 5% WB | MCDBR | | **9.2** | **11.8** | **11.8** | **13.9** | **15.6** | **13.7** | **12.1** | **11.9** | **11.8** | **9.7** | **6.3** | **7.9** | (38) |
| (39) | | | MCWBR | | **9.1** | **10.9** | **9.7** | **9.3** | **9.5** | **7.8** | **6.7** | **7.1** | **7.7** | **7.2** | **5.9** | **7.8** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.230** | **0.262** | **0.271** | **0.333** | **0.340** | **0.354** | **0.403** | **0.358** | **0.326** | **0.285** | **0.259** | **0.212** | (40) |
| (41) | | taud | | | **2.208** | **2.218** | **2.271** | **2.221** | **2.400** | **2.407** | **2.322** | **2.442** | **2.514** | **2.545** | **2.246** | **2.159** | (41) |
| (42) | | Ebn at Noon | | | **696** | **809** | **902** | **882** | **900** | **888** | **831** | **852** | **836** | **795** | **639** | **618** | (42) |
| (43) | | Edh at Noon | | | **52** | **78** | **96** | **120** | **107** | **108** | **116** | **96** | **78** | **58** | **50** | **41** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **0.81** | **1.91** | **3.55** | **5.26** | **5.56** | **5.82** | **5.40** | **4.23** | **2.82** | **1.44** | **0.69** | **0.51** | (44) |
| (45) | | RadStd | | | **0.06** | **0.14** | **0.28** | **0.44** | **0.46** | **0.56** | **0.35** | **0.40** | **0.27** | **0.19** | **0.07** | **0.06** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (1 neighbor) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page